

Project STORM and Beyond, All Purpose Modular Flight Computer

GENERAL DESCRIPTION

The MONAD system is the culmination of many years of ANU Rocketry electronics design techniques, fulfilling all current requirements for flight computers for whole team project vehicles. It features the following broad components:

- STM32 microcontroller with high performance F4 arm cortex
- Spartan 6 xc6slx16 FPGA with 16K LCs, DSP, 400MHZ io and 256Mb external RAM
- OV7670 VGA Camera
- Quectel Cellular/GPS module, Supports LTE CAT M1, NB2, GNSS, Class 5 transmitter
- Matek MR900 MLRS Transceiver with 1W output
- 3-axis gyroscope, 3-axis accelerometer, 3-axis compass, with onboard DMP, multiple temperature sensors and a high precision barometer

MONAD supports significant auxiliary device interfaces including but not limited to hardware I²C and USART, Analog signal processing, External charging interface, Multiple toggleable 3.3V and 5V power lines, 13 Full customizable FPGA interface pins.

RENDER

APPLICATIONS

- Complete flight computer system
- Control many peripheral devices
- General purpose communications

FEATURES

- USB C fast charging at up to 5A
- Triple fan active cooling, individually articulable
- Dual redundant MicroSD slots
- Built in programming interfaces
- 25 pin GPIO connector
- Onboard Digital Motion Processing (DMP)
- Hot swappable battery
- Battery heater circuit for in flight thermal management
- External 4 port UART debugging probe
- RoHS and Green compliant

TYPICAL OPERATING DIAGRAM

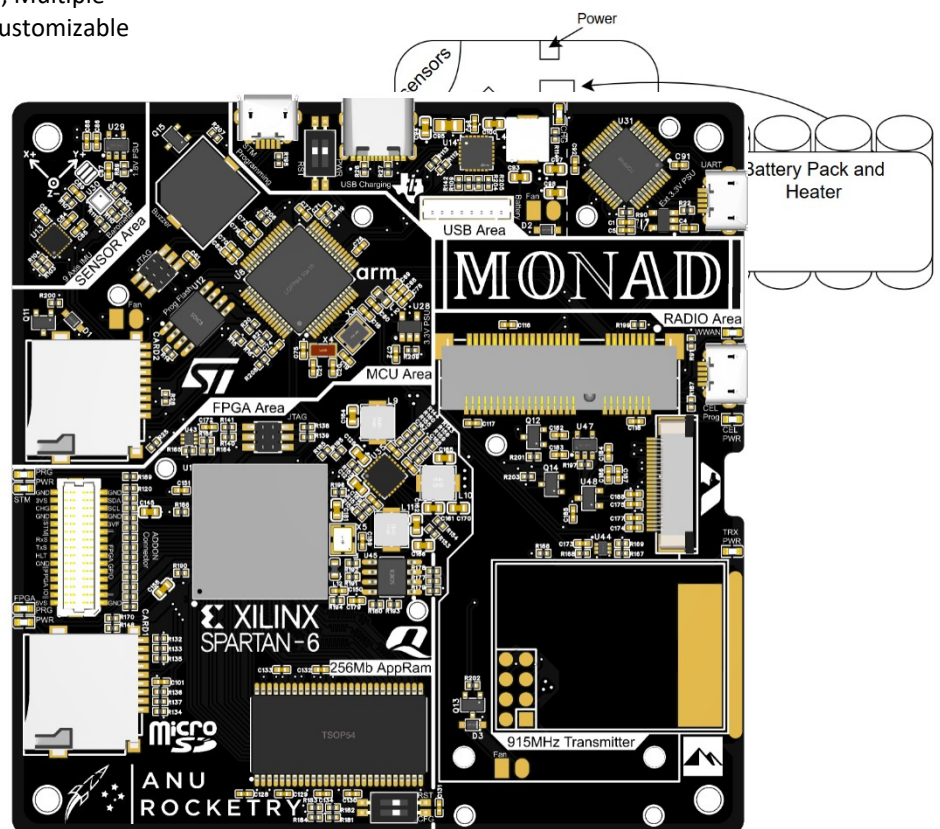


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1 GENERAL DESCRIPTION

1.1 PURPOSE AND SCOPE

This document is a preliminary data sheet, providing a description, specifications, and design related information on the MONAD flight computer.

For further details relating to individual components, please refer to the MONAD component map and Component Descriptions.

1.2 PRODUCT OVERVIEW

The MONAD Flight Computer is a multi-module PCB (MMP) comprised of two core systems integrated together into a single unified package. One system is a traditional Microcontroller (MCU) STM32 F405, and is responsible for general purpose data logging, in-flight operations, telemetry transmission and system power management. It interfaces with all on-board sensors, including the 9-axis IMU ICM-20948, BMP580 barometer, TMP102 thermometers, as well as primary power supply ics, the BQ25895 charger and EA3056 step down supply.

The MCU also shares an I2C and USART bus with the FPGA system

The second system is the Field Programmable Gate Array (FPGA) and related peripherals. The FPGA is responsible for bulk and high-speed interfacing, including image processing, storing and transmission, External peripheral interfacing and acts as a true redundant alternative for core functions of the MCU.

MONAD features a rich, optimized suite of sensors, and allows users to further customize via peripherals, fulfilling the minimum requirement without overcomplicating design and manufacturing.

The included sensors are:

- ICM20948 9 Axis IMU
- BMP580 Barometer
- TMP102 I2C temperature sensors x2
- Additionally, 4 further temperature sensors are present, 3 through inclusion in onboard devices, and one onboard the normally attached battery heater

MONAD, with its high integration, complete feature set, and extreme modularity, enables users to eliminate the costly and complex selection, qualification, and system level integration of compute devices and peripherals, guaranteeing a simplified user experience.

1.3 APPLICATIONS

- Complete flight computer system
- Interface for many peripheral devices within a vehicle
- General purpose communications hub
- Datalogger

2 DEVICE FEATURES

2.1 STM32F405 FEATURES

The STM32F405 Microcontroller includes the following features:

- Core: Arm® 32-bit Cortex®-M4 CPU with FPU, operating at up to 168 MHz, delivering 210 DMIPS with DSP instructions.
- 1 MB of Flash memory and 192 KB of SRAM, including 64 KB of core-coupled memory (CCM) data RAM.
- External 128Mbit program flash and SD card slot.
- Many desirable connectivity options: USB OTG, Ethernet, ADCs, camera interface.
- Several power modes, including wake on interrupt.

2.2 XC6SLX16 FEATURES

The XC6SLX16 FPGA includes the following features:

- 14,579 logic cells.
- 576 Kbits of block RAM, 256Mb external SDRAM.
- 32 DSP slices for efficient signal processing.
- 232 I/O pins, 13+ broken out for easy access.
- Low-power, 45nm process technology.

2.3 MATEK MR900 FEATURES

The MATEK MR900 within the MONAD mainboard includes a wide range of features:

- STM32 running mLRS firmware.
- SX1262 with TCXO for stable frequency performance.
- Up to 30dBm (1000mW) for long-range communication.
- Operates at 900MHz (supports 868MHz and 915MHz bands).
- Bi-directional Serial MAVLink, CRSF or SBUS RC outputs.

2.4 BG95-M3 mPCIe FEATURES

The BG95-M3 mPCIe includes the following capabilities:

- Maximum downlink of 588 Kbps and uplink of 1119 Kbps.
- Standard Mini PCIe form factor measuring 51.0mm × 30.0mm × 4.9mm.
- Supports multiple frequency bands for worldwide deployment.
- Optimized for IoT applications requiring efficient power usage.

2.5 OV7670 FEATURES

The OV7670 includes the following features:

- 640x480 (VGA) with support for various output formats including RGB565, YUV, and Raw RGB data.
- Standard SCCB interface is compatible with I2C protocol.
- Capable of 30 frames per second for VGA resolution.
- Includes functions like auto white balance, auto gain control, and auto exposure control.
- Low power consumption of 60mW.

3 ELECTRICAL CHARACTERISTICS

3.1 D.C. Electrical Characteristics

Typical Operating Circuit of section 4.2, VDD = 1.8V, VDDIO = 1.8V, T_A=25°C, unless otherwise noted.

PARAMETER	CONDITIONS	MIN	TYP	MAX	UNITS	NOTES
SUPPLY VOLTAGES						
VDD		1.71	1.8	3.6	V	1
VDDIO		1.71	1.8	1.95	V	1
SUPPLY CURRENTS						
9-Axis (DMP disabled)	Low-Noise Mode; Compass in Continuous Mode		3.11		mA	2
Gyroscope Only (DMP, Barometer & Accelerometer disabled)	Low-Power Mode, 102.3 Hz update rate, 1x averaging filter		1.23		mA	2
Accelerometer Only (DMP, Barometer & Gyroscope disabled)	Low-Power Mode, 102.3 Hz update rate, 1x averaging filter		68.9		μA	2
Magnetometer Only (DMP, Accelerometer & Gyroscope disabled)	8 Hz update rate		90		μA	2
Full-Chip Sleep Mode			8		μA	2
TEMPERATURE RANGE						
Specified Temperature Range	Performance parameters are not applicable beyond Specified Temperature Range	-40		+85	°C	1

Table 4. D.C. Electrical Characteristics

NOTES:

1. Derived from validation or characterization of parts, not guaranteed in production.

3.2 Other Electrical Characteristics

Typical Operating Circuit of section 4.2, VDD = 1.8V, VDDIO = 1.8V, T_A=25°C, unless otherwise noted.

PARAMETER	CONDITIONS	MIN	TYP	MAX	UNITS	NOTES
SERIAL INTERFACE						
SPI Operating Frequency, All Registers Read/Write	Low Speed Characterization		100 ±10%		kHz	
	High Speed Characterization		7 ±10%		MHz	
I ² C Operating Frequency	All registers, Fast-mode			400	kHz	
	All registers, Standard-mode			100	kHz	

Table 6. Other Electrical Specifications

NOTES:

1. Derived from validation or characterization of parts, not guaranteed in production.

3.3 ABSOLUTE MAXIMUM RATINGS

Stress above those listed as “Absolute Maximum Ratings” may cause permanent damage to the device. These are stress ratings only and functional operation of the device at these conditions is not implied. Exposure to the absolute maximum ratings conditions for extended periods may affect device reliability.

PARAMETER	RATING
Supply Voltage, VBUS, CHG	-2V to +22V
Battery Voltage, BAT+	3V to 4.2V
REGOUT	-0.5V to 2V
Input Voltage Level (SCL, SDA, Tx, Rx, MCU IO)	-0.5V to 3V3 + 0.5V
Input Voltage Level (FPGA IO)	-0.5V to D3V3 + 0.5V
Input Voltage Level (USB micro)	-2V to + 6V
Acceleration (Any Axis, unpowered)	200g for 0.2 ms
Operating Temperature Range	-40°C to +70°C
Storage Temperature Range	0°C to +30°C
Electrostatic Discharge (ESD) Protection	2kV (HBM); 200V (MM)

Table 9. Absolute Maximum Ratings

4 SYSTEM INFORMATION

4.1 PIN OUT AND SIGNAL DESCRIPTION FOR STM32

Pin ID	Pin ID 2	Pin ID 2	Pin No.	Function
PA0			14	Cellular (USART 2) CTS
PA1			15	Cellular (USART 2) RTS
PA2			16	Cellular (USART 2) Tx
PA3			17	Cellular (USART 2) Rx
PA6	A2	OTG	22	Pull high to enable 5v power system
PA7	A3	INT	23	Receives active low, 256- μ s pulse to host to report charger device status and fault.
PA11	D-		44	USB Data minus for programming and interfacing
PA12	D+		45	USB Data plus for programming and interfacing
PA13	SWDIO		46	JTAG programming Data IO
PA14	SWCLK		49	JTAG programming Clock
PA15	FLASH_CS		50	SPI flash CS pin
PC0	D8_HEATER		8	Battery Heater Enable
PC1	D13		9	Programmable LED
PC2	D12		10	MLRS Fan Enable
PC3	D11		11	Charger Fan Enable
PC4	A4		24	User IO, Analog capable
PC5	A5		25	User IO, Analog capable
PC6	D6		37	FPGA (USART 6) Tx
PC7	D5		38	FPGA (USART 6) Rx
PC8	SDIO_D0		39	Micro SD Data 0
PC9	SDIO_D1		40	Micro SD Data 1
PC10	SDIO_D2		51	Micro SD Data 2
PC11	SDIO_D3		52	Micro SD Data 3
PC12	SDIO_CLK		53	Micro SD Clock
PC13			3	RTC Crystal X1
PC14			4	RTC Crystal X2

Pin ID	Pin ID 2	Pin ID 2	Pin No.	Function
PB0	BUZZER		26	Buzzer Enable
PB3	FLASH_SCK		55	SPI flash SCK pin
PB4	FLASH_MISO		56	SPI flash MISO pin
PB5	FLASH_MOSI		57	SPI flash MOSI pin
PB6	SCL		58	I2C Clock Line
PB7	SDA		59	I2C Data Line
PB8	D9		61	PSU Fan Enable
PB9	D10		62	FPGA/External UART Halt pin (pullup default)
PB10	TX_D1		29	MLRS (USART 3) Tx
PB11	RX_D0		30	MLRS (USART 3) Rx
PB12	SD_DETECT		33	Micro SD Card Detect
PD2	SDIO_CMD		54	Micro SD CMD
PH0			5	Main Crystal X1
PH1			6	Main Crystal X2
BOOT0				Enter boot mode (JTAG)
NRST	RESET			Resets the device (Switch provided)

Table 11. Pinout Descriptions

4.2 I2C DEVICE ADDRESSES

Binary Address	Hex Address	Component
110101	0x6A	EA3056 Multi Step Down
1000110	0x8C	BMP580 Barometric Sensor
1001000	0x90	FPGA Adjacent Temperature Sensor TMP102
1001001	0x92	MLRS Adjacent Temperature Sensor TMP102
1001010	0x94	Battery Adjacent Temperature Sensor TMP102
1010000	0xA0	UART USB Multiplexer XR21V1414
1101000	0xD0	IMU ICM-20948
1101010	0xD4	Charger IC BQ25895
N/A	N/A	FPGA (Master or Slave, user determinable)

Table 11. I2C Addresses

5 APPLICATION INFORMATION

5.1 PIN OUT DIAGRAM AND SIGNAL DESCRIPTION FOR GPIO CONNECTOR

PIN NUMBER	PIN NAME	PIN DESCRIPTION
1, 12, 23, 24, 29, 30	GND	Digital Ground Global
2	5VS	System 5V line, STM can toggle in software, powers MR900 also.
3-11, 13, 15, 17, 19	FPGA GPIO	FPGA GPIO pins, current limited, See below for pin IDs.
14	UART HLT	UART halt, dedicated pin for multi device UART negotiation.
16	TxS	STM32 UART Transmit pin.
18	RxS	STM32 UART Receive pin.
20,22	A4, A5	STM32 GPIO pins.
21	D3F	FPGA linked 3.3v line, will toggle with the FPGA to save power.
25	SCL	Main I2C clock, access onboard peripherals or connect your own.
26	CHG	Feed 5v - 18v to charge battery. Solar compatible!
27	SDA	Main I2C data, access onboard peripherals or connect your own.
28	3VS	STM32 linked 3.3v line, always on.

Table 10. Pinout Descriptions

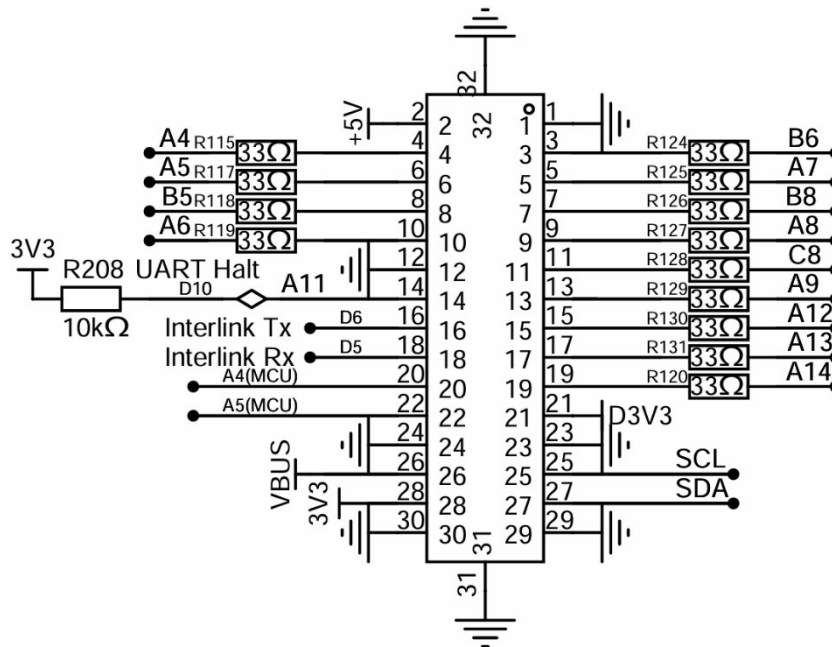


Figure 3. Pin out Diagram for 30 pin GPIO Header

5.2 MR900-30 MLRS COMMANDS

Command	Description
help, h, ?	Displays the help page.
v	Prints device and version information.
pl	Lists all parameters.
pl c	Lists common parameters.
pl tx	Lists Tx parameters.
pl rx	Lists Rx parameters.
p name	Gets the value of a parameter.
p name = value	Sets the value of a parameter.
p name = ?	Gets the value of a parameter and allowed values.
pstore	Stores updated parameters to memory.
setconfigid	Select the configuration ID.
bind	Start binding process.
reload	Reload all parameter settings.
stats	Start streaming statistics.
listfreqs	List frequencies used in the FHSS scheme.
systemboot	Call the system bootloader.
hc04 pt	Enter serial passthrough mode.
hc04 setpin	Set the pin of the HC04 module.

Table 11. MLRS CLI COMMANDS

NOTES:

1. Add underscores instead of spaces when typing command names out, as an example, P RX_Power = 0;.
2. Best practice is to add a semicolon at the end of commands to signify line termination.